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LEADER & ACHIEVER COURSE

PHASE : MLA,B,C,H,I,J,K,M,P,Q,R,S,T,U,V & MAZA,B,C,D,E,F,K,P,Q,R,X & MAZY

TARGET : PRE-MEDICAL 2024

Test Type : **SRG TEST** Test Pattern : **NEET (UG)**

TEST DATE : 10-08-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	4	3	1	2	3	4	1	1	1	1	4	1	4	3	3	2	1	2	3	3	2	3	1	4	2	2	3	2	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50										
A.	2	3	1	3	1	3	3	4	3	2	4	3	3	2	2	2	2	2	1	2										

HINT – SHEET

SUBJECT : CHEMISTRY

SECTION-A

1. Ans (1)

$$2.3 \times 10^{-5}\ s^{-1} \times 90 \times 60s = 2.303 \log\left(\frac{A_0}{A_t}\right)$$

$$\log\left(\frac{A_0}{A_t}\right) = 0.054 \Rightarrow [A_t = 0.885\ A_0]$$

2. Ans (4)

$$\frac{K_2}{K_1} = \frac{\ell n 25}{\ell n 2} = 4.65$$

3. Ans (3)

(Step-2) $R = K[CHCl_3]\,[Cl]$

(Step-1) $K_{eq} = \frac{[Cl]^2}{[Cl_2]} \Rightarrow [Cl] = (K_{eq}[Cl_2])^{1/2}$

$\therefore\ R = (K\,k_{eq})\,[CHCl_3]\,[Cl_2]^{1/2}$

4. Ans (1)

Zero order $\Rightarrow t_{50\%} = \frac{[A_0]}{2K}$

$$\log(t_{50\%}) = \frac{1}{2K} \times \log[A]_0$$

5. Ans (2)

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6. Ans (3)

	A	⟶	nB
t = 0	A_0		0
t = t	$A_0 - x$		nx

$(A_0 - x) = nx$

$A_0 = nx + x$

$$x = \frac{A_0}{(n+1)}$$

$$[B] = nx = \frac{n}{(n+1)}A_0$$

7. Ans (4)

$A \xrightarrow{K_I} P$ (I^{st} order)

$B \xrightarrow{K_{II}} Q$ (zero order)

$$\frac{K_I}{K_{II}} = \frac{\frac{2.303}{30\ min}\log\left(\frac{1}{0.5}\right)}{\frac{(2-1)M}{30\ min}} = 0.693$$

8. Ans (1)

[A]	[B]	[C]	rate
0.02	0.02	0.02	2.08×10^{-3}
0.01	0.02	0.02	1.04×10^{-3}
0.02	0.04	0.02	4.16×10^{-3}
0.02	0.02	0.04	8.32×10^{-3}

rate $\propto [A]^1\,[B]^1\,[C]^2$

rate $= K\,[A]\,[B]\,[C]^2$

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9. **Ans (1)**

$$\log K = \log A - \frac{Ea}{2.303R} \times \left(\frac{1}{T}\right)$$

$$2.9 = \log A - \frac{Ea}{2.303R} \times (1.3 \times 10^{-3}) \quad \ldots(1)$$

$$1.1 = \log A - \frac{Ea}{2.303R} \times (1.5 \times 10^{-3}) \quad \ldots(2)$$

from eq (1) & (2)

$Ea \approx 4 \times 10^4$ cal/mol

10. **Ans (1)**

Zero order $\Rightarrow R = K$

half order $\Rightarrow R = K[A]^{1/2}$

2nd order $\Rightarrow t_{1/2} \propto \frac{1}{[A_0]}$

I^{st} order $\Rightarrow t_{1/2} = \frac{0.693}{K}$

11. **Ans (1)**

$R = K[A]^\circ = K$

& K is independent of concentration of reactant & irrespective of order R & K depend upon temperature.

12. **Ans (4)**

For gaseous compound ; pressure = concentration × RT

$\Rightarrow$ Pressure $\propto$ concentration

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14. **Ans (4)**

Activation energy is less than threshold energy.

Boltzman factor $\propto \frac{1}{e^{Ea/RT}}$

all radioactive disintegration are of 1st order.

$H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$ explosive reaction.

15. **Ans (3)**

Unit of K = $mol^{1-n}\, L^{n-1}\, S^{-1}$

order fractional

& upon dilution $\Rightarrow$ concentration decreases

$\Rightarrow$ Rate decreases

16. **Ans (3)**

1. Molecularity is integer only
2. Instantaneous rate is measured at an instant of time
3. $(t_{1/2})_{n=0} \propto [A]$
4. Rate law = Mathematical expression representing concentration of reactant.

17. **Ans (2)**

Fact

18. **Ans (1)**

Catalyst doesn't change Gibbs free energy.

20. **Ans (3)**

(A) is correct but (R) is incorrect

21. **Ans (3)**

Since, the rate law contains $[H^+]$ term. This means that rate changes with variation in the value of pH.

22. **Ans (2)**

For elementary reaction, $M \longrightarrow N$

Rate law can be written as rate $\propto [M]^n$

$Rate = k[M]^n$ ...(i)

When we double the concentration of [M], rate becomes 8 times, hence, new rate law can be written as

$$8 \times rate = k[2M]^n \Rightarrow \frac{Rate}{8 \times rate} = \frac{k[M]^n}{k[2M]^n}$$

$$\therefore \quad \frac{1}{8} = \frac{1}{[2]^n} \Rightarrow 2^n = 8 = 2^3 \Rightarrow n = 3$$

23. **Ans (3)**

$$R = \frac{-d[H^+]}{dt} = \frac{0.01M}{10^{-12}s} = 1 \times 10^{10} M/s$$

24. **Ans (1)**

$(t_{1/2})$ is independent of Reactant's pressure $\Rightarrow$ Reaction follow's I^{st} order.

25. **Ans (4)**

2 Step reaction followed by intermediate.

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26. Ans (2)

$R = k[A]^1$

$1 \times 10^{-3} = k[0.1\ M]$

$k = 10^{-2}\ s^{-1}$

or $2 \times 10^{-3} = K\ [0.2\ M]$

$K = 10^{-2}\ s^{-1}$

or $3 \times 10^{-3} = K\ [0.3\ M]$

$K = 10^{-2}\ s^{-1}$

27. Ans (2)

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28. Ans (3)

NCERT XII Pg. # 105, Edition - 2019

29. Ans (2)

NCERT XII Pg. # 117, Edition - 2019

30. Ans (2)

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31. Ans (2)

NCERT XII Pg. # 104, 106, Edition - 2013

32. Ans (3)

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33. Ans (1)

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34. Ans (3)

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SECTION-B

36. Ans (3)

$K_a = A\, e^{-\frac{E_a}{RT}}$

$K_b = A\, e^{-\frac{E_a}{RT}}$

$$\frac{K_a}{K_b} = \frac{e^{-\frac{E_a}{RT}}}{e^{-\frac{E_b}{RT}}} = e^{\left(\frac{-E_a}{RT}+\frac{E_b}{RT}\right)} = e^{\frac{(E_b-E_a)}{RT}}$$

Temp. tends to $T \longrightarrow \infty$

$$\frac{E_b - E_a}{RT} \xrightarrow[\text{to}]{\text{tends}} 0$$

$$\frac{K_a}{K_b} = e^o = 1$$

37. Ans (3)

$\because\ t_{1/2} = \frac{1}{aK}$

$$\log\ t_{1/2} = \log\left(\frac{1}{a}\right) + \log\frac{1}{K}$$

$$\log\ t_{1/2} = -\log a + \log\frac{1}{K}$$

38. Ans (4)

$$2A + B \xrightarrow{K} C + D$$

at t = 0 Co 2Co 0 0

at t = 30 min Co−2x 2Co-x x x

at 30 min [C] $= \frac{Co}{4} = x$

$[A] = [Co-2x] = Co - \frac{2Co}{4} = \frac{2Co}{4}$

$[B] = [2Co-x] = 2Co - \frac{Co}{4} = \frac{7Co}{4}$

$r = k[A]\,[B]^2$

$$= k \times \frac{2Co}{4} \times \left(\frac{7Co}{4}\right)^2 = 49K\frac{Co^3}{32}$$

39. Ans (3)

$A \longrightarrow$ Product, $[A_t] = \frac{A_0}{2^n} = \frac{4}{2^3} = 0.5M$

$R \longrightarrow$ Product, $t_{100\%} = 2t_{1/2} = 40$ min

$[A_t]$ at 60 min = 0

40. Ans (2)

$$NH_4NO_2(aq) \longrightarrow N_2(g) + 2H_2O(\ell)$$

ni a 0 0

nt a − x x 2x

n_∞ 0 a 2a

a = 70 & x = 40

a − x = 30

$$k = \frac{2.303}{1200}\log\frac{70}{30}\sec^{-1}$$

$$= \frac{2.303}{1200}\log\frac{7}{3}\sec^{-1}$$

41. Ans (4)

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42. Ans (3)

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43. Ans (3)

Flask I

1 Lit of 1 M solution of A

$\downarrow$

0.5 M $\quad t_{½}$ ⟲ $t = 8$ hrs.

$\downarrow$

0.25 M $\quad t_{½}$

$\therefore \; 2t_{1/2} = 8$

$t_{1/2} = 4$

Flask II

100 ml of 0.6 M solution of A

$\downarrow \quad t_{1/2} = ?$

0.3 M

$t_{1/2} = 4$ hrs.

[$\because$ for Ist order reaction $t_{1/2}$ does not depends on concentration]

44. Ans (2)

$E_a = RT$

$K = Ae^{-RT/RT}$

$K = Ae^{-1}$

$K = \frac{A}{e} = 0.37\ A$

$K = 37\%$ of A

45. Ans (2)

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46. Ans (2)

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$nA \longrightarrow B$

$r = -\frac{1}{n}\frac{dA}{dt} = 0.01$

$-\frac{1}{n} - \frac{0.8}{40} = 0.01$

$\frac{1}{n}(0.02) = 0.01$

$n = 2$

47. Ans (2)

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48. Ans (2)

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$A \longrightarrow 2B$		
a	0	a = 0.4M
a – x	2x	2x = 0.6
(0.4–0.3)	0.6	x = 0.3
0.1 M	0.6	

$k = \frac{2.3}{20}\log\frac{0.4}{0.1} = \frac{2.3 \times 0.6}{20} = 0.069\ sec^{-1}$

$r = k\,[A]^1$

$= 0.069 \times 0.4 = 0.0276\ M\ sec^{-1}$

49. Ans (1)

$r = k'\,[CH_3COOCH_3]\,[H_2O]$

$r = k[CH_3COOCH_3]$

$k = k'\,[H_2O]$

$k' = \frac{k}{[H_2O]} = \frac{\frac{2.3}{20}\log\frac{0.8}{0.2}}{[55.55]}$

$k' = \frac{\left(\frac{2.3\times0.6}{20}\right)}{55.55} = \frac{0.0693}{55.55}(mole^{-1}L^{+1}sec^{-1})$

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